Zachary Land

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Education

Imperial College London Sep 2024 — Jul 2028

- **Course**: Joint Mathematics and Computer Science
- **Awarded Grade**: 82.7% average; expected 1st class
- **Modules**: Analysis, Multivariable Calculus, Numerical Linear Algebra, Programming, Statistical Modelling

Hereford Sixth Form College & Whitecross Hereford High School Sep 2017 — Jul 2024

- **Courses**: Computer Science, Further Mathematics, Mathematics, Physics
- **Awarded Grades**: A*A*A*A*; 10 grade 9 GCSEs

Experience

SC26 International Student Cluster Competition Jul 2026 — Present

- Selected to represent the UK for the International Student Cluster Competition taking place in November.
- Optimising matrix multiplication, NAMD, and LINPACK for faster execution ahead of the competition.

Google Research Engineer Intern Jun 2026 — Sep 2026

- Working on classifier probes and embedding systems for LLM content safety.
- Researching the effects of long contexts on current methods for content safety.

Aon Risk Modeller Intern Jun 2025 — Aug 2025

- Implemented a statistical model to forecast catastrophic losses and analyse market risk.
- Self-taught VBA to innovate Excel macros, reducing data science pipeline execution time by 10 minutes.

Projects

vLLM KV Cache Quantization May 2026 — Jun 2026

- Expanded LLM context length 4x by quantizing vLLM KV cache from FP16 to blockwise NVFP4.
- Preserved vLLM paging efficiency and used QAT distillation to quantize the transformer to FP4.

ICHack Winner Feb 2026

- Won ICHack, Europe's largest student-run hackathon, with a mathematical model to predict oil spill trajectories.
- Formulated ODEs and monte-carlo simulations on a graph lattice to calculate routes with minimal ecological risk.

Metaphor Alignment Research Paper Feb 2026 — Apr 2026

- Authored a CCN-accepted extended abstract on detecting conceptual metaphor alignment in embedding spaces.
- Developed a graph-based metric that identified metaphor alignments missed by CKA.

UCL AI Research Hackathon Winner Nov 2025 — Jul 2026

- Presented a machine learning model with representational alignment to identify discrepancies in LLM output.
- Researching an extension of the project in collaboration with, and funded by, Holistic AI.

Probabilistic Computing Research Oct 2025 — Jan 2026

- Built a probabilistic numerical linear algebra package and an Ising model to expedite OPM-MEG brain scanning.
- Awarded best undergraduate research project in Oxford's probabilistic computing research competition.

IBM-Z Datathon Winner Oct 2025

- Competed against 6500+ participants to develop a real-time riot prediction model.
- Applied NLP with PCA for sentiment analysis and a GNN to forecast unrest through influence networks.

Skills

Programming Languages: Python, C++, C, SQL, VBA, C#, Java, Kotlin, Haskell
Languages: English (native), Spanish (A2), Japanese & Mandarin (elementary)

Achievements & Extracurricular

Gold in the British Physics Computational Challenge; Gold in the British Physics Olympiad at the age of 16
Top 30 in the UK for Long Jump at the age of 14; V6-Level Climber